Xiao Lin

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Research Interests: Embodied Foundation Models & World Models, MLLM, Agentic Memory, 3D Perception

EDUCATION

Tongji University, *Ph.D. Candidate in Control Science and Engineering* | Shanghai, China Sep. 2021 – Mar. 2027

- **Research**: Focused on Embodied AI, MLLM, Agentic Memory, 3D Perception.

Tongji University, *B.S. in Automation* | Shanghai, China Sep. 2017 – Jun. 2021

- **Reward**: **National Scholarship** (2020) | Excellent Graduation Project (2021) | Outstanding Graduate (2021)

MAIN INTERNSHIP EXPERIENCE

JD Future Academy Jun. 2026 – Present

Algorithm Research Intern, Advisors: Tong He, Jiangmiao Pang

Research Topic: Embodied Foundation Models & World Models

- Contributed to post-training stages of **JoyBrain**, unifying 2D/3D/video grounding data and training pipelines. Built and maintained **10M+** multimodal samples and evaluated across **11** benchmarks, raising the overall ER score from 53.1 to **64.8** with **28.0/13.5 pp** gains in 3D/2D grounding.
- Co-trained VQA and discrete actions autoregressively with **19M** action samples, improving the RoboCasa score from 47.00 to **50.41** (**+3.41 pp**).
- Trained a unified **MoT** world model on large-scale interleaved language–image planning data, outperforming BAGEL-7B-MoT by **35%** on joint planning benchmarks and Cosmos3-Nano on world-state prediction.

Alibaba DAMO Academy Jan. 2026 – Jun. 2026

Algorithm Research Intern, Advisors: Xin Li, Deli Zhao

Research Topic: Embodied Foundation Models

- Engineered native 3D grounding for **RynnBrain 1.1** [1, 2] using million-scale, multi-source 3D data, achieving **23.44 AP3D** on WildDet3D-Bench and **41.12 AP@15** on SUN RGB-D.
- Developed the CoT-driven embodied planning framework **RynnBrain-Plan** (8B / 30B-A3B MoE) for long-horizon planning via SFT, and **RynnBrain-CoP** via Chain-of-Point reasoning and two-stage SFT + RL (**GRPO**) alignment, achieving **11%** gains over Qwen3-VL-8B on RefSpatial-Bench and **7%** gains on complex spatiotemporal task.

China Telecom AI Institute (TeleAI) Sep. 2025 – Jan. 2026

Algorithm Research Intern, Advisors: Dell Zhang, Xuelong Li

Research Topic: Memory Systems / Agentic Retriveal

- Open-sourced **TeleMem** [5] (490+ GitHub stars), a RAG-based long-term memory system using vector-retrieval for Agentic AI, improving accuracy by **19%** and cutting token costs by **43%** on ZH-4O benchmark. Published two papers.
- Architected a multi-level memory system supporting hour-long videos, achieving a **4.7**× processing speedup.
- Implemented RL-based **Search Agent** with Verl for **multi-step retrieval** and **tool use**, improving Video-MME accuracy by **23%** and reducing average retrieval rounds from **5 to 3**.

ADDITIONAL INTERNSHIP EXPERIENCE

Huawei Cloud | **Pangu Vision Foundation Model** Oct. 2024 – Mar. 2025

Algorithm Research Intern, Advisor: Dongsheng Jiang

Research Topic: Vision Foundation Models / Multimodal LLMs

- Engineered an MAE-based sparse self-supervised pre-training paradigm for vision foundation models, scaling models to **30B** parameters on a **48-GPU H20** distributed cluster.
- Reduced computational complexity from quadratic to linear through structural optimizations, achieving a **1.5**× training speedup.

SELECTED PUBLICATIONS

[1] *RynnBrain 1.1: Towards More Capable and Generalizable Embodied Foundation Model*. **Technical Report**, **Core Contributor**, 2026. arXiv | Code

[2] *RynnBrain: Open Embodied Foundation Models*. **Technical Report**, **Contributor**, 2026. arXiv | Code

[3] **Xiao Lin** *et al. MemPose: Category-level Object Pose Estimation with Memory*. **ECCV**, 2026. arXiv

[4] **Xiao Lin** *et al. CleanPose: Category-level object pose estimation via causal learning and knowledge distillation*. **ICCV**, 2025. arXiv | Code

[5] **Xiao Lin*** *et al. TeleMem: Building Long-Term and Multimodal Memory for Agentic AI*. **Tech Report**. arXiv | Code

[6] **Xiao Lin** *et al. CLIPose: Category-Level Object Pose Estimation with Pre-trained Vision-Language Knowledge*. **IEEE TCSVT**, 2024. arXiv

[7] **Xiao Lin** *et al. TransPose: 6D Object Pose Estimation with Geometry-aware Transformer*. **Neurocomputing**, 2024. arXiv

[8] Minghao Zhu, **Xiao Lin** *et al. Fine-Grained Spatiotemporal Motion Alignment for Contrastive Video Representation Learning*. **ACM MM**, 2023. arXiv | Code

[9] Minghao Zhu, **Xiao Lin** *et al. TACO: Towards Task-Consistent Open-Vocabulary Adaptation in Video Recognition*. **NeurIPS 2026**. arXiv

[10] Yun Peng, **Xiao Lin** *et al. VLLM-LAD: Visual Large Language Model for Zero-shot Logical Anomaly Detection*. **IEEE TCSVT**, 2025. Paper

[11] Yun Peng, **Xiao Lin** *et al. SAM-LAD: Segment Anything Model Meets Zero-Shot Logic Anomaly Detection*. **Knowledge-Based Systems (KBS)**, 2025. arXiv

SELECTED RESEARCH EXPERIENCE

Open-World 3D Perception & World Modeling Jul. 2022 – Present

Primary Contributor

- **CleanPose** (Accepted at **ICCV 2025**) [4]: Proposed a **causal learning** framework to mitigate data bias and improve robustness to unseen distributions.
- **MemPose** (Accepted at **ECCV 2026**) [3]: Developed a category-level pose estimation framework with a memory buffer, bridging single-frame perception and multi-view physical consistency.
- **CLIPose** (Published in **IEEE TCSVT 2024**) [6]: Leveraged pretrained CLIP knowledge through multimodal contrastive learning to align image, point-cloud, and text representations.

ACADEMIC SERVICE

Reviewer: ICCV, ECCV, CVPR, NeurIPS, ICML, ACM MM, IEEE TCSVT, IEEE TMM, IEEE TASE.